

301 TXOne Solutions

Basic Implementation Training Labs

Network Defense Solutions

Rev 3.3_250326

Global Training Services Team
TXOne Network Inc

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Revision History

Version	Author	Date	Change
v2.1	Louis Liu	2023/02/15	1 st edition
v2.1b	Louis Liu	2023/03/05	2 nd edition, 2 attendees share 1 EdgeIPS
v2.1c	Louis Liu	2023/04/18	3 rd edition. Added Edge Lab 8: EdgeFire NAT settings
v2.2c	Louis Liu	2023/05/15	Using TXPI for MP installation
v2.2c	Chris Van Den Abbeele	2023/06/04	Minor cosmetic updates
v2.3	Louis Liu	2023/06/05	- EdgeOne Installation & Configuration - EdgeIPS 103 Installation & Configuration - Portable Inspector
v2.4	Louis Liu	2023/06/13	Split lab guide by solutions based on v2.3
v3.0	Louis Liu	2024/04/05	Win7 VM consolidation
v3.2	Chris Van Den Abbeele	2025/01/29	Revised for improved phrasing
v3.3	Louis Liu	2025/02/05	EdgeOne 2.4 , Edge device firmware 2.1
v3.3	Louis Liu	2025/03/25	Update for EdgeIPS Pro

Product Versions

Solutions	Edition
EdgeOne	2.2.20, 2.4.18
EdgeIPS Pro 216102	2.1.35

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1. Lab scenarios mapped to product features

Table 1 Lab scenarios mapped to product features

Solution	Lab scenario
EdgeOne	Lab1: EdgeOne Deployment 1. Installation, basic CLI commands
	Lab2: Initial setup / upgrade of EdgeOne and web UI walk thru 1. Administration functions: License, System Time, Updates, Account Management, Proxy 2. EdgeOne upgrade to the new version
EdgeIPS Pro 216	Lab3: EdgeIPS Pro 216 Hardware, Installation and lab setup 1. DIN Rail Mounting 2. Power Connection 3. VMWare network configuration for lab exercises
	Lab4: EdgeIPS Pro 216 device initialization 1. Device settings 2. Operation mode 3. LED Status
	Lab5: Register EdgeIPS Pro 216 with EdgeOne for centralized management 1. Device System time setting 2. EdgeIPS Pro 216 sync to EdgeOne using passcode authentication method
	Lab6: Device Group Management 1. Device Groups 2. Group Settings
	Lab7: Policy Enforcement Configuration 1. Policy Enforcement operation modes 2. Basic Filter 3. Advanced Filter

2. Lab Environment content list

2.1 Virtual Machines

Table 2 Virtual Machines

Host	Server Name ZIP Filename	VMware Image Name	Hardware Specification	Purpose
A	LabConsole_xx xxx.zip	LabConsole_x xxx	CPU: 2 Core RAM: 4GB Hard Disk: 100G Network: NAT / Bridged (e.g. to VMnet11)	Main Console for labs
A	EdgeOne	edgeone_x.x. x.ova	CPU: 8 Core RAM: 16GB Hard Disk: 25G/80G Network: NAT / Bridged (e.g. to VMnet11)	EdgeOne server
B	Win7_xxxx.zip	Win7_xxxx	CPU: 2 Core RAM: 2GB Hard Disk: 60G Network: NAT / Bridged (e.g. to VMnet11)	Target for EdgeIPS Pro 216,

2.2 Hardware/Software Requirements

Table 3 Hardware / Software Requirements

No	Items	Specification			
1	Laptop	OS	Windows 10 / 11		
		Hard Drive	65GB minimum		
		Memory	16GB minimum, 24GB Recommend		
		Network Adapter	WIFI NIC for Internet Access LAN NIC for Lab practice		
		Hard drive space requirements			
		Solutions	Space Requirements	Memory Requirements	VMs
		EdgeOne EdgeIPS Pro 216	Host A: 48GB Host B: 20GB	Host A: 4.5GB Host B: 1GB	Host A: - LabConsole - EdgeOne Host B: - Win7
2	VMWare Workstation Pro 17	Hypervisor Minimum Hardware level = 16			
3	EdgeIPS Pro 216	Model IPSP-216-BP-TM (2 attendees share 1 EdgeIPS Pro 216 device)			
4	Network Cable	2 UTP Network cables (Each attendee)			
5	Network Switch	5 ports model			

2.3 Lab network architecture

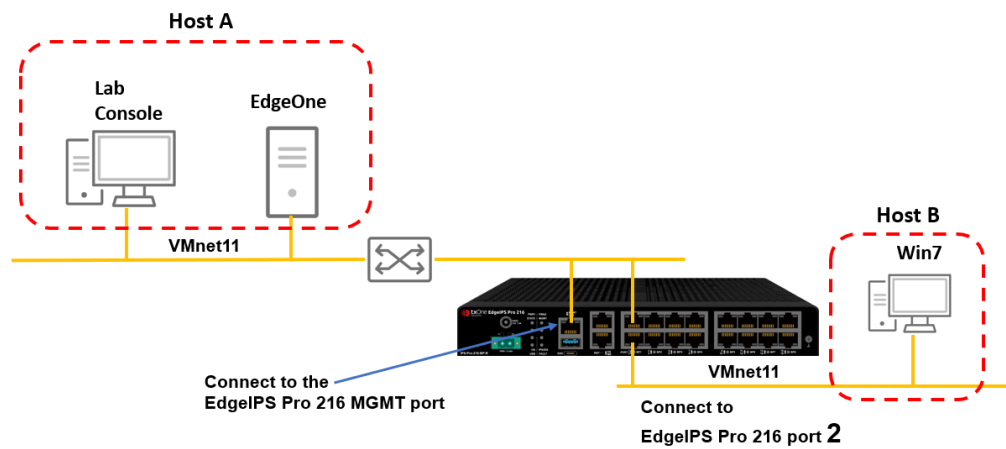


Figure 1 Lab network architecture

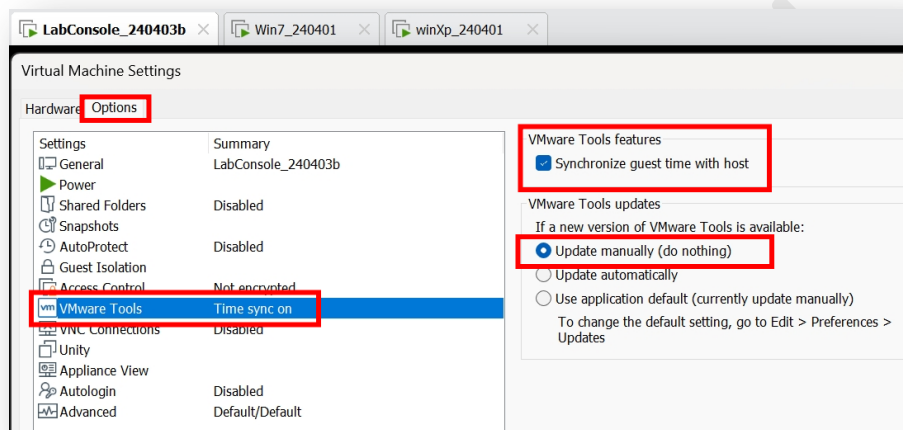
2.4 Credentials

Table 4 Lab Credentials

No	VM Image	IP	Account / Password
1	LabConsole	192.168.23.196	txone/12345678
2	EdgeOne vx.x	192.168.23.21	Web: txone/12345678 Server: root/12345678
3	Win7	192.168.23.24	txone/12345678
4	EdgelPS Pro 216	192.168.23.205	txone/12345678

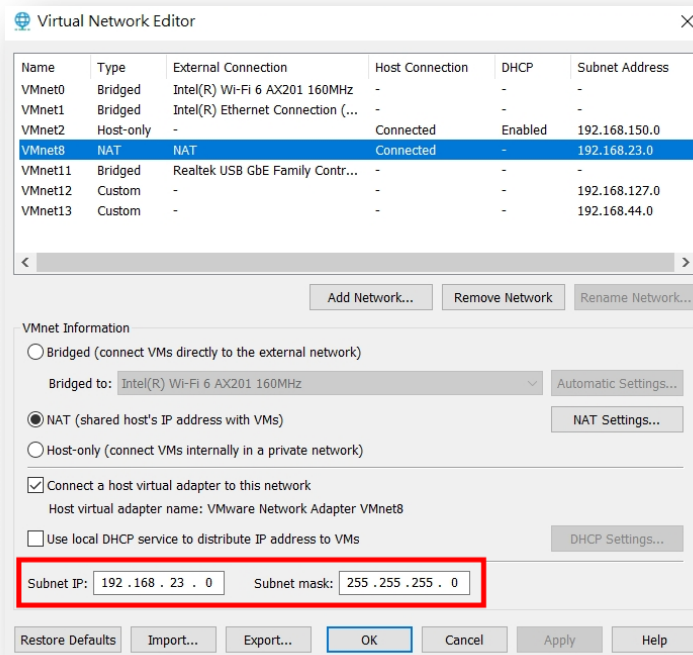
2.5 VMWare VMTools update Settings

- Edit the Settings of each VM
- Choose the **Options** tab
- Select the **VMWare Tools** option
- Check **“Synchronize guest time with host”**
- Under VM Tools updates, select the option **“Update manually (do nothing)”**

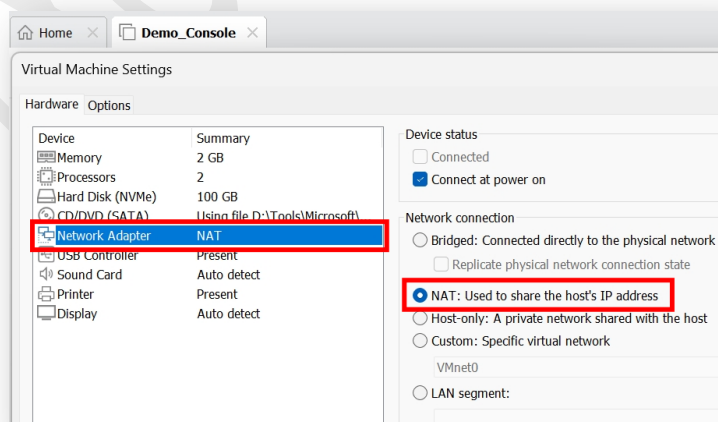


2.6 VMWare Networks Settings

- Make sure your laptop can access the Internet via its wireless interface. You will also need a UTP network interface, which will be used to connect to the EdgeIPS Pro 216.
- In VMware workstation, open the Virtual Network Editor and configure the VMWare NAT network to 192.168.23.0 /24.



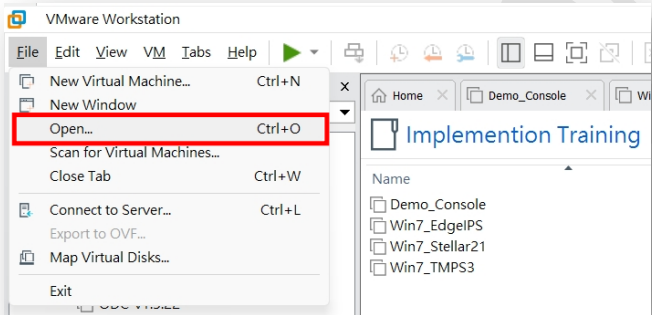
- Set the Network Adapter to NAT for LabConsole.
 - The default NAT gateway setting is **192.168.23.2**.
 - LabConsole and EdgeOne use **8.8.8.8** as DNS server.

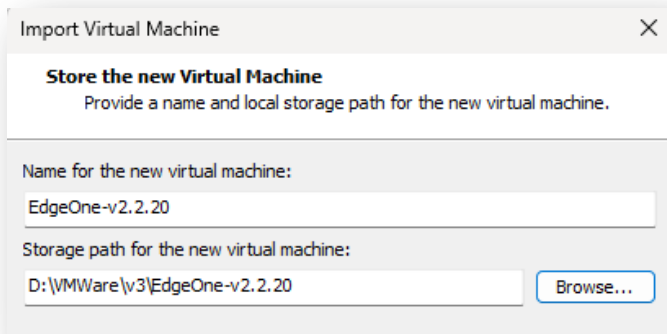
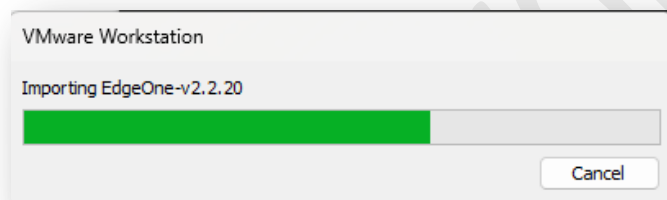
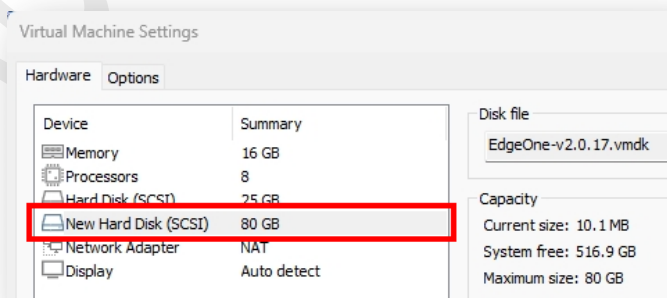


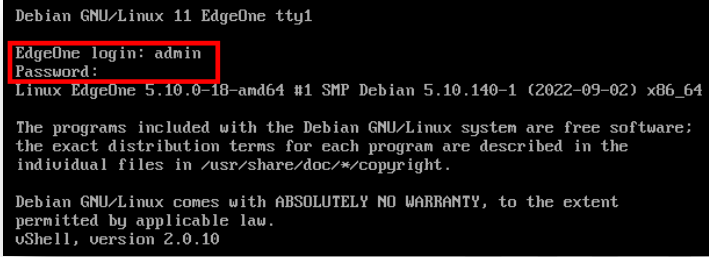
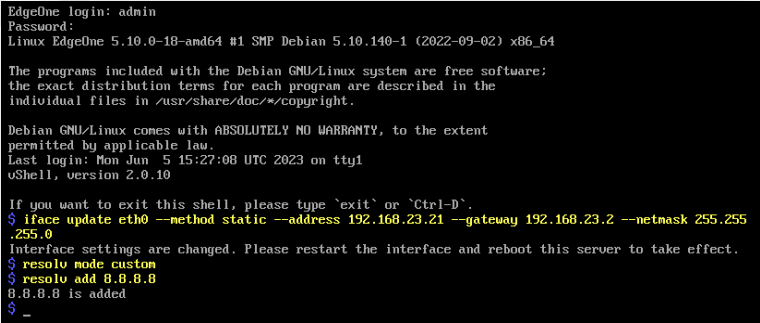
3. Labs

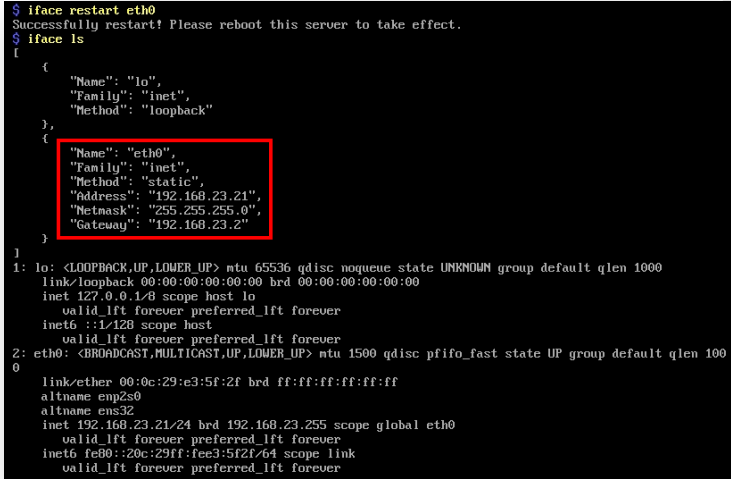
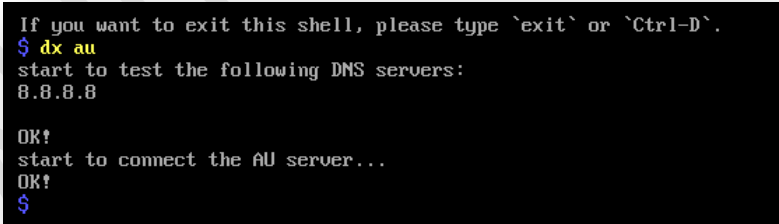
Lab1: EdgeOne Deployment

Table 5 EdgeOne Deployment

Items	Descriptions
Application target	Host A
Steps	Import the EdgeOne OVA VMWare Image - Launch VMWare Workstation and open the EdgeOne OVA image.  - Select the OVA image file “edgeone_xxx.ova” from Training package.  - Name the Virtual Machine “EdgeOne-vxxx” and select Storage Path for the new virtual machine. Note: If you are using OneDrive on your host, ensure that the storage path of the VM is not a directory synchronized with OneDrive. Every time the OneDrive agent syncs with the server, it can disrupt the integrity of the virtual

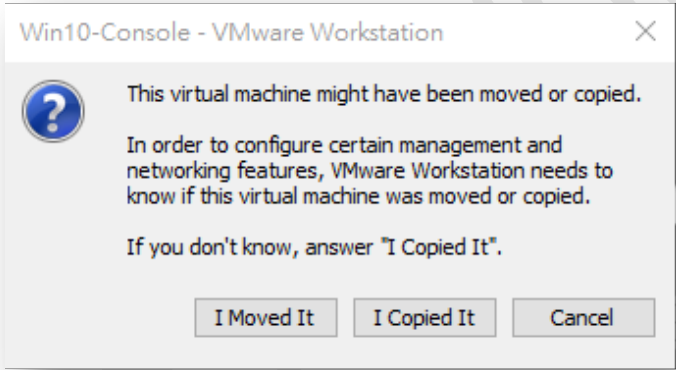
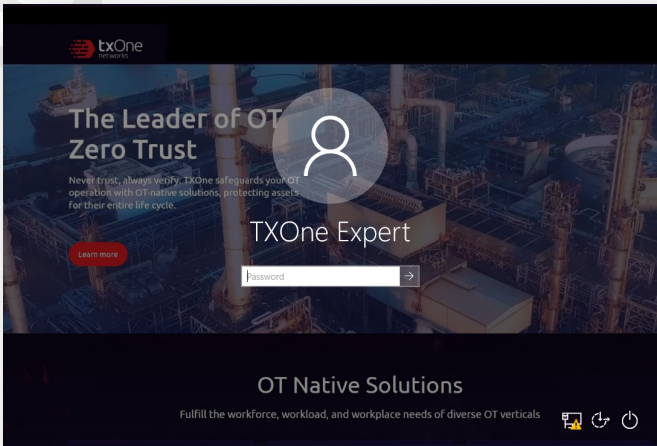
Items	Descriptions
	machine, leading to numerous errors in the EdgeOne CLI.  4. The EdgeOne OVA image import process will take a moment.  5. After the OVA image import has completed. Add the 2nd hard drive to the EdgeOne virtual machine. Allocate at least 80GB for the 2nd hard drive (SCSI Disk type). Do not worry if you do not have that space on your laptop. This hard disk is “thin provisioned”. This means it will only consume the space that it needs, but it needs to be configured as 80GB 

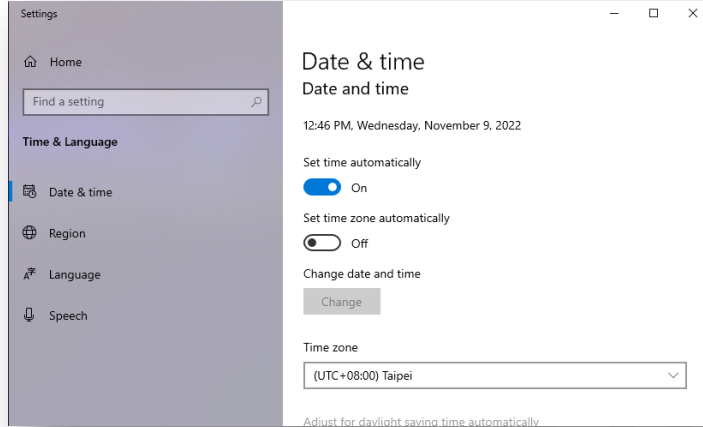
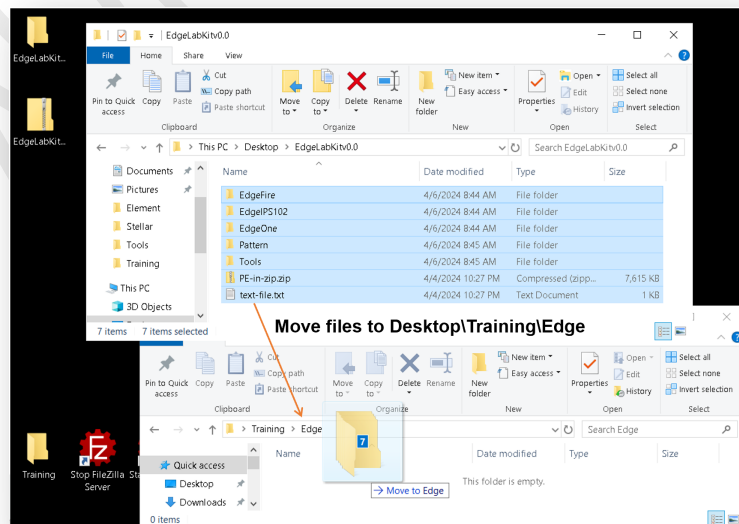
Items	Descriptions
	EdgeOne Server Network configuration - Make sure that the VMWare Network Adapter is set to NAT and turn on the EdgeOne virtual machine. Log in to the EdgeOne web UI with the following credentials. User: admin Password: txone  - Run the “oobe” command to change the root password to “12345678”. - Use the new credentials “admin/12345678” to Log in to the EdgeOne server again. - Run the following commands in the restricted oobe shell to configure the network interface: iface update eth0 --method static --address 192.168.23.21 --gateway 192.168.23.2 --netmask 255.255.255.0 - Setup DNS resolv mode custom resolv add 8.8.8.8 

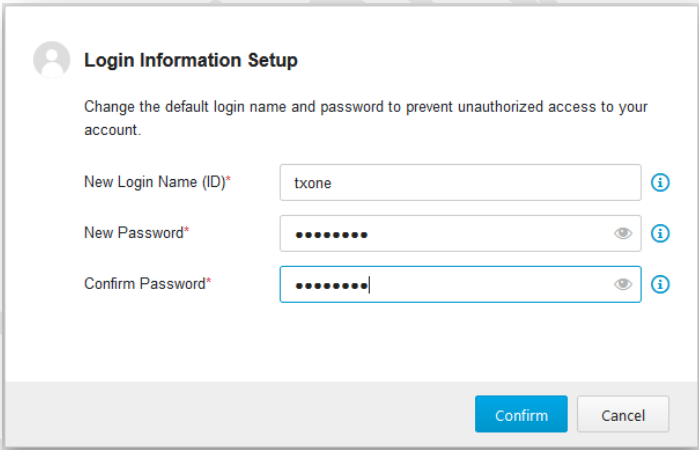
Items	Descriptions
	6. Check the network settings and restart the EdgeOne virtual machine: <pre>iface restart eth0</pre> iface ls reboot 7. Note: if it takes a long time to execute the “iface restart eth0” command, then please check your IP address settings.  <pre>\$ iface restart eth0 Successfully restart! Please reboot this server to take effect. \$ iface ls { { "Name": "lo", "Family": "inet", "Method": "loopback" }, { "Name": "eth0", "Family": "inet", "Method": "static", "Address": "192.168.23.21", "Netmask": "255.255.255.0", "Gateway": "192.168.23.2" } } 1: lo: <LOOPBACK,UP,LOWER_UP> mtu 65536 qdisc noqueue state UNKNOWN group default qlen 1000 link/loopback 00:00:00:00:00:00 brd 00:00:00:00:00:00 inet 127.0.0.1/8 scope host lo valid_lft forever preferred_lft forever inet6 ::1/128 scope host valid_lft forever preferred_lft forever 2: eth0: <BROADCAST,MULTICAST,UP,LOWER_UP> mtu 1500 qdisc pfifo_fast state UP group default qlen 1000 link/ether 00:0c:29:c3:5f:2f brd ff:ff:ff:ff:ff:ff altname emp2s0 altname ens32 inet 192.168.23.21/24 brd 192.168.23.255 scope global eth0 valid_lft forever preferred_lft forever inet6 fe80::20c:29ff:fe3:5f2f/64 scope link valid_lft forever preferred_lft forever</pre> 8. Log in to EdgeOne and enter the “dx au” command to the test the connection with update server. The command will show the result of the connection to the DNS server and the AU server.  <pre>If you want to exit this shell, please type `exit` or `Ctrl-D`. \$ dx au start to test the following DNS servers: 8.8.8.8 OK! start to connect the AU server... OK! \$</pre>
Expected Result	EdgeOne server imported and running up successful.

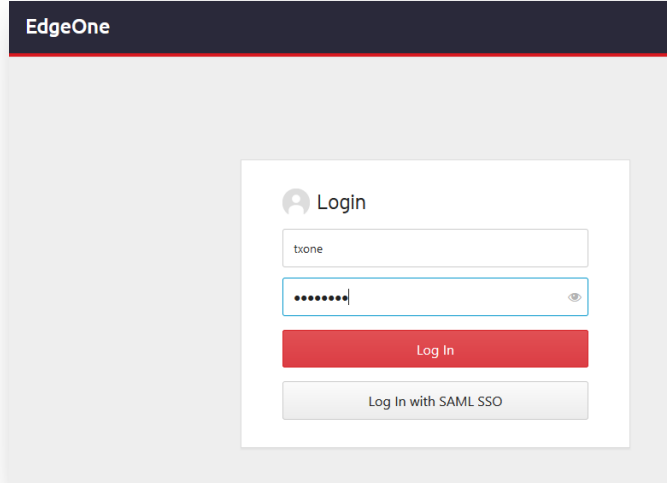
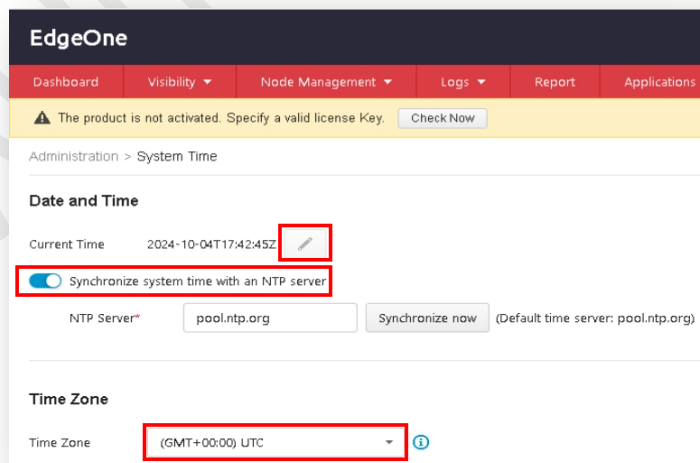
Lab2: Initial setup / upgrade of EdgeOne and web UI walk thru

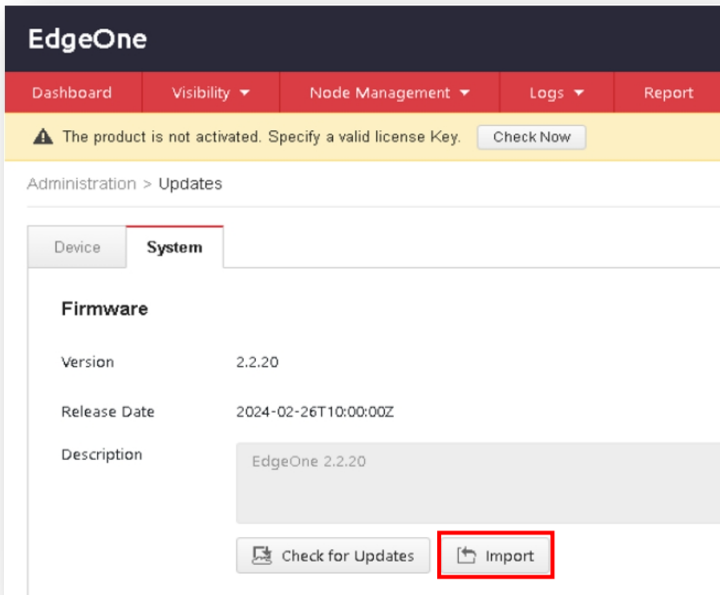
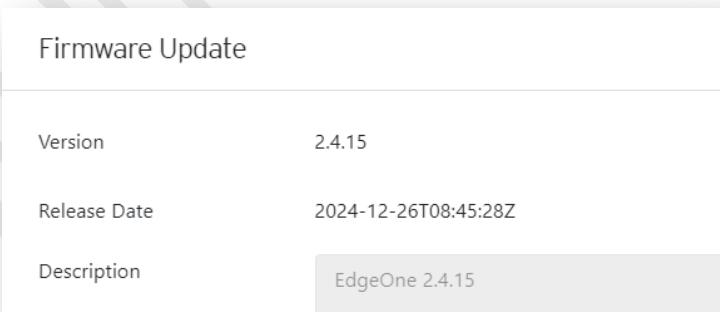
Table 6 Initial setup / upgrade of EdgeOne and web UI walk thru

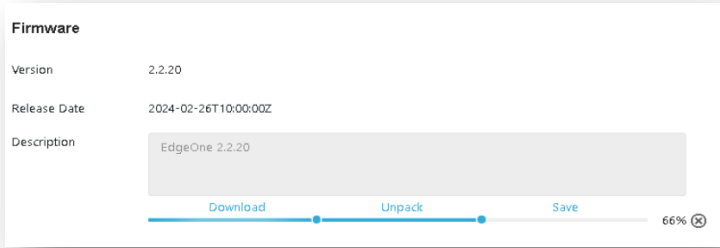
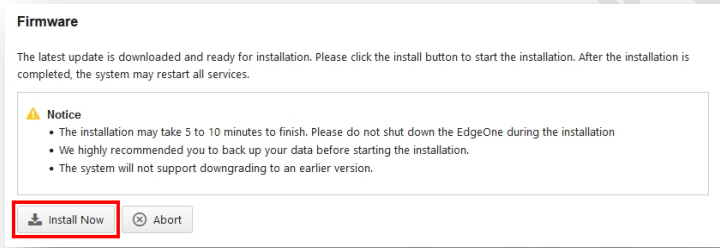
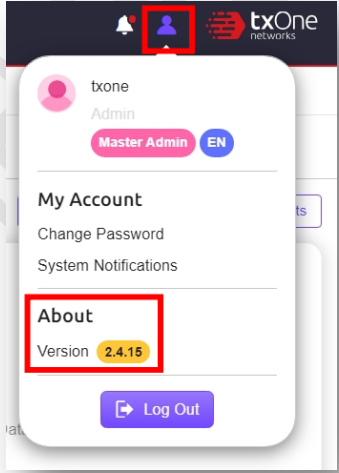
Items	Descriptions
Application target	Host A
Steps	Configure access to the EdgeOne web UI from the VM LabConsole 1. In VMWare Workstation, open the LabConsole VM. 2. Revert VM LabConsole to snapshot "ReadyforLabs" 3. Set the VM LabConsole network adapter to NAT and Turn on the VM. 4. if the VM is powered on for the first time, then click "I Moved It"  5. Log in using the following credentials. 6. User: txone (TXOne Expert) 7. Password: 12345678 

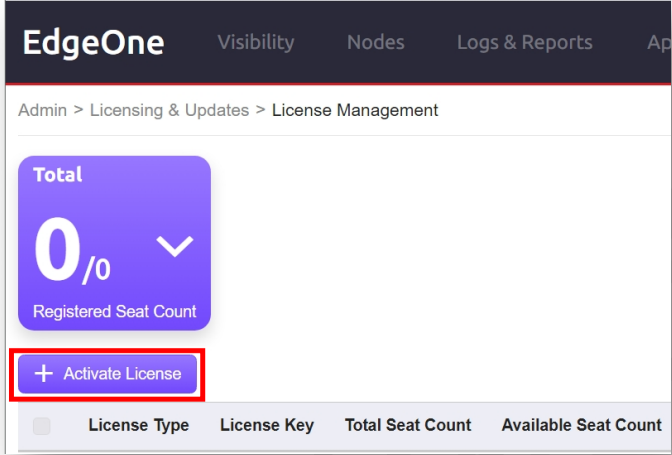
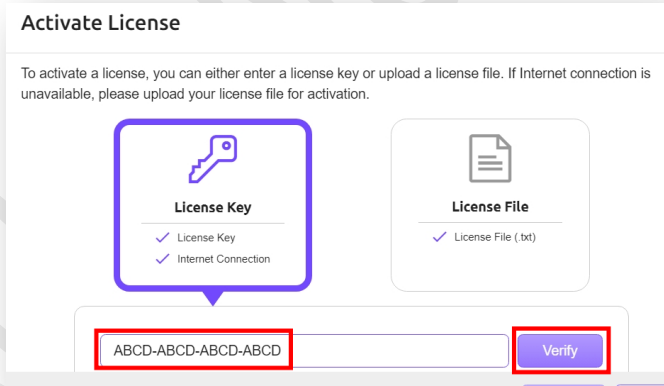
Items	Descriptions
	8. Configure the Windows OS Time Zone for the LabConsole VM 9. For the purpose of this training, set the time zone to UTC  10. Copy the “EdgeLabKitVx.y.zip” that was shared with you to the desktop of the LabConsole and expand it (right click the file and “Extract All”). It will create the “EdgeLabKitVx.y.zip” folder on the desktop. 11. Move all subdirectories from that folder to the Edge folder which is in the Training Folder on the Desktop, as shown in the screenshot below. 12. The training folder is hosted on a webserver running in this VM. We will access it from other VMs to download the required files. 13. Move the files “PE-in.zip” and “text-file.txt” from \Desktop\Training\Edge to \Desktop\Training 

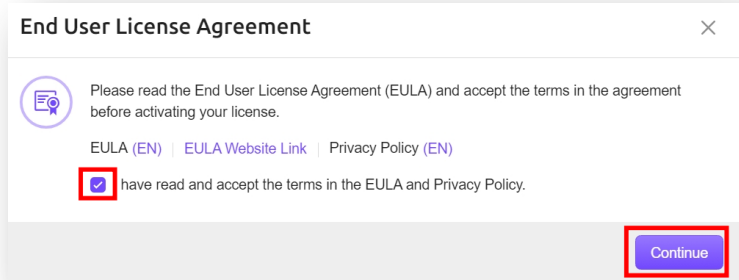
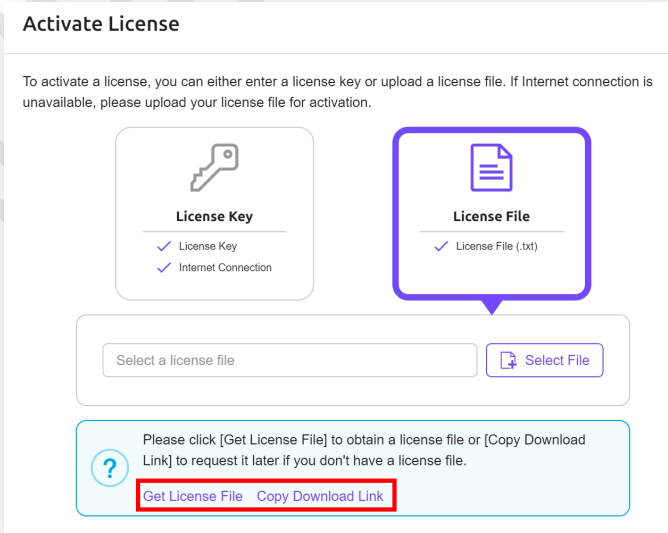
Items	Descriptions
	14. Launch the Chrome browser from the Windows taskbar and go to: https://192.168.23.21. 15. Login as admin / txone 16. Click “Confirm” to accept the EULA and OT Intelligent Trust.  The dialog box is titled "End User License Agreement (EULA) and OT Intelligent Trust". It contains two checked checkboxes: "I understand that I must accept the End User License Agreement (EULA) before activating the license later." and "I agree to enable OT Intelligent Trust (recommended)". A blue "Continue" button is at the bottom right. 17. You will be prompted to create a new user and set password.  The dialog box is titled "Login Information Setup". It instructs the user to "Change the default login name and password to prevent unauthorized access to your account." It contains three input fields: "New Login Name (ID)*" with the value "txone", "New Password*" with masked characters, and "Confirm Password*" with masked characters. Each field has an information icon. At the bottom right are "Confirm" and "Cancel" buttons.

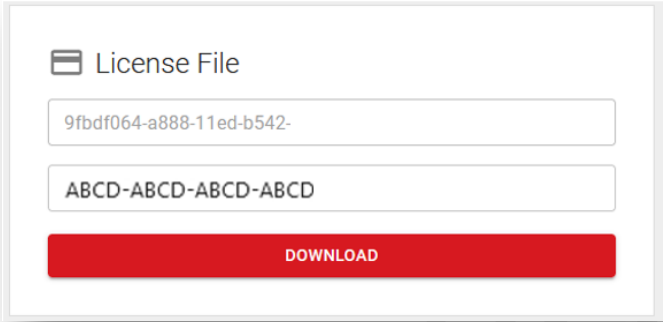
Items	Descriptions
	18. Create the user "txone" with the password "12345678," then log back into the EdgeOne web UI.  System Time setting - Go to Administration > System Time - Set date and time manually or synchronize with an NTP server. - For our labs, it's OK to keep the time zone to UTC Note: For an actual deployment, the time zone should be set to match your region. 

Items	Descriptions
	Upgrade EdgeOne - Go to Administrator > Updates, select the “System” tab and click on “Import”. - Choose the EdgeOne update firmware file from the folder Desktop\Training\Edge\EdgeOne. It has the acf extension.  - EdgeOne will check the integrity of the firmware. 

Items	Descriptions
	4. EdgeOne will show the process stages: Download, Unpack and Save.  5. Click “Install Now”. Normally, the update will take less than 1 minute.  6. Go to Log in User > About and check the version of EdgeOne. 

Items	Descriptions
	Activate a license using the “online” method. - In EdgeOne go to Admin > Licensing & Updates > License Management and click “+Activate License”.  - Click “Online license activation”. Enter the EdgeOne Node license key that will be provided by your trainer and click “Verify”. 

Items	Descriptions
	3. Accept the End User License Agreement  The dialog box titled "End User License Agreement" contains a message: "Please read the End User License Agreement (EULA) and accept the terms in the agreement before activating your license." Below this are links for "EULA (EN)", "EULA Website Link", and "Privacy Policy (EN)". A checkbox is checked and labeled "I have read and accept the terms in the EULA and Privacy Policy." A "Continue" button is at the bottom right. Activate a license using the "License File" method: 4. For air gapped EdgeOnes, Offline license activation can be used. Let's use this option to add the licenses for EdgeIPS Pro 216 Choose "License File" then click "Get License File" or "Copy Download link" depending on your Internet connection. <u>Get License File:</u> Choose this option if your EdgeOne web UI has Internet access. <u>Copy Download Link:</u> If your EdgeOne web UI has no Internet access, copy this link to a client that can access the internet.  The "Activate License" dialog box provides instructions: "To activate a license, you can either enter a license key or upload a license file. If Internet connection is unavailable, please upload your license file for activation." It shows two options: "License Key" (with sub-points "License Key" and "Internet Connection") and "License File" (with sub-point "License File (.txt)"). The "License File" option is highlighted with a blue box. Below is a "Select a license file" input field with a "Select File" button. At the bottom, a message says: "Please click [Get License File] to obtain a license file or [Copy Download Link] to request it later if you don't have a license file." The "Get License File" and "Copy Download Link" buttons are highlighted with a red box.

Items	Descriptions
	5. EdgeOne in our lab environment is connected via NAT to your host and thus has internet access. So click “Get License File”. Enter an EdgeIPS Pro 216 software license key and download the license file. In an air-gapped environment, you would copy the download link to a computer with internet access, to get the license file from there.  6. Return to EdgeOne web UI, select License File and click “Apply” to upload the license file to EdgeOne. 